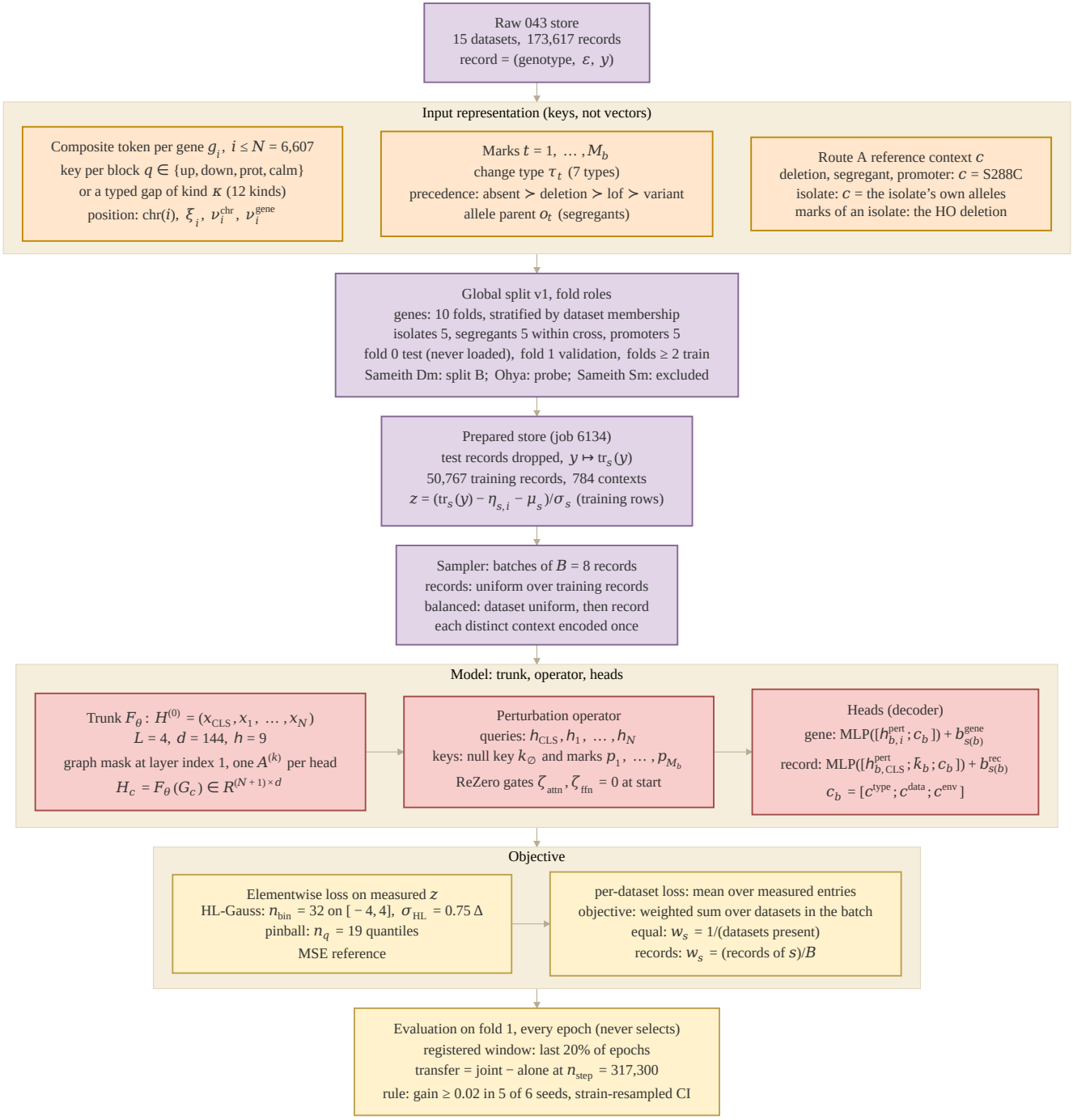

Raw 043 store
15 datasets, 173,617 records
record = (genotype, ε, y)
Input representation (keys, not vectors)
Composite token per gene g_i, $i \le N = 6{,}607$
key per block $q \in \{\text{up}, \text{down}, \text{prot}, \text{calm}\}$
or a typed gap of kind κ (12 kinds)
position: chr(i), ξ_i, ν_i^{chr}, ν_i^{gene}
Marks $t = 1, \ldots, M_b$
change type τ_t (7 types)
precedence: absent $\succ$ deletion $\succ$ lof $\succ$ variant
allele parent o_t (segregants)
Route A reference context c
deletion, segregant, promoter: c = S288C
isolate: c = the isolate's own alleles
marks of an isolate: the HO deletion
Global split v1, fold roles
genes: 10 folds, stratified by dataset membership
isolates 5, segregants 5 within cross, promoters 5
fold 0 test (never loaded), fold 1 validation, folds ≥ 2 train
Sameith Dm: split B; Ohya: probe; Sameith Sm: excluded
Prepared store (job 6134)
test records dropped, $y \mapsto \text{tr}_s(y)$
50,767 training records, 784 contexts
$z = (\text{tr}_s(y) - \eta_{s,i} - \mu_s)/\sigma_s$ (training rows)
Sampler: batches of $B = 8$ records
records: uniform over training records
balanced: dataset uniform, then record
each distinct context encoded once
Model: trunk, operator, heads
Trunk F_θ : $H^{(0)} = (x_{\text{CLS}}, x_1, \ldots, x_N)$
$L = 4$, $d = 144$, $h = 9$
graph mask at layer index 1, one $A^{(k)}$ per head
$H_c = F_\theta(G_c) \in R^{(N+1)\times d}$
Perturbation operator
queries: $h_{\text{CLS}}, h_1, \ldots, h_N$
keys: null key $k_\varnothing$ and marks $p_1, \ldots, p_{M_b}$
ReZero gates $\zeta_{\text{attn}}, \zeta_{\text{ffn}} = 0$ at start
Heads (decoder)
gene: MLP($[h_{b,i}^{\text{pert}}; c_b]$) + $b_{s(b)}^{\text{gene}}$
record: MLP($[h_{b,\text{CLS}}^{\text{pert}}; \bar{k}_b; c_b]$) + $b_{s(b)}^{\text{rec}}$
$c_b = [c^{\text{type}}; c^{\text{data}}; c^{\text{env}}]$
Objective
Elementwise loss on measured z
HL-Gauss: $n_{\text{bin}} = 32$ on $[-4, 4]$, $\sigma_{\text{HL}} = 0.75\,\Delta$
pinball: $n_q = 19$ quantiles
MSE reference
per-dataset loss: mean over measured entries
objective: weighted sum over datasets in the batch
equal: $w_s = 1/(\text{datasets present})$
records: $w_s = (\text{records of } s)/B$
Evaluation on fold 1, every epoch (never selects)
registered window: last 20% of epochs
transfer = joint − alone at $n_{\text{step}} = 317{,}300$
rule: gain ≥ 0.02 in 5 of 6 seeds, strain-resampled CI